Exercise - 2 Solutions

2.1 Safety Vocabulary

Define the following terms in your own words and give a concrete example from an autonomous driving context for each:
1. Loss
2. Hazard
3. Risk

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Loss - the actual bad outcome, something that has already happened and caused harm. In Autonomous Vehicle: collision with pedestrian will lead to loss of life.

Hazard - A system state that could lead to a loss if conditions are bad enough. The bad outcome hasn't happened yet but the system is in a dangerous configuration. In an Autonomous Vehicle: it is moving while a pedestrian is in its path.

Risk - How much we should worry. It combines how likely the hazard is, how likely it leads to loss if it occurs, and how bad that loss would be.
Risk ≈ P(hazard) × P(loss | hazard) × Severity.
In AV: the risk of striking a pedestrian is high if the vehicle operates at night (high P(hazard)) without a fallback (high P(loss|hazard)) at highway speed (high severity).

NOTE:The key distinction: a hazard is not yet a loss. A car moving near a pedestrian is a hazard. The collision is the loss.

2.2 ODD Specification

Based only on the system description in the course Roadmap (you have not yet seen the training data), define the Operational Design Domain (ODD) for the CARLA system.
Cover at least the following dimensions: weather, lighting, camera condition, scene type, and vehicle speed.
For each dimension, state:
1. The operating conditions (where the system is considered valid).
2. The non-operating conditions (where it is not).
3. How an ODD violation could be detected at runtime

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The key insight: the ODD defines the "contract" under which safety claims are valid. Everything outside it is undefined territory.

Dimension	Operating (valid)	Non-operating (invalid)	Runtime detection
Weather	Clear, cloudy, no precipitation	Rain, fog, snow, hail	Image brightness variance drops; dedicated weather classifier
Lighting	Daytime with normal sunlight	Night, dusk/dawn, tunnels, glare	Mean pixel luminance below threshold
Camera	Forward-facing, clean, fixed mount	Obstructed, dirty, tilted, wet lens	Blur score / sharpness metric on incoming frame
Scene type	Urban roads with structured lanes	Highways, off-road, construction zones	GPS geofence + lane structure detection
Vehicle speed	Low speed / intersection approach (< ~50 km/h)	High-speed driving	Onboard speedometer reading

2.3 Losses

STPA begins by identifying the losses that must be prevented - unacceptable outcomes involving harm to people, property, or the environment.
For each loss, identify its ID (L-1, L-2, ...), briefly describe it, and explain why it is unacceptable:

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ID	Loss description	Why unacceptable
L-1	Injury or death of a pedestrian	Direct harm to human life, highest possible severity
L-2	Injury or death of vehicle occupants	System is responsible for passenger safety
L-3	Collision with another vehicle	Property damage, potential injury, legal liability

L-4	Running a red light / violating traffic law	Endangers others; triggers legal consequences and loss of operating license
L-5	Loss of system trust / regulatory shutdown	Systemic harm, entire technology grounded, halting safety benefits at scale

2.4 Hazards

A hazard is a system state that, combined with worst-case environmental conditions, can lead to a loss. Hazards are properties of the system, not of individual components.
1. Identify at least four system-level hazards (H-1, H-2, ...) for the CARLA system.
2. For each hazard, state which loss(es) from Exercise 2.3 it can lead to.
3. Rate each hazard's likelihood (low / medium / high) and its severity (low / medium / high).

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ID	Hazard	Loss(es)	Likelihood	Severity
H-1	Pedestrian in vehicle path while vehicle is moving without braking	L-1, L-2	Medium	High
H-2	Vehicle continues through intersection with undetected traffic signal	L-3, L-4	Medium	High
H-3	Vehicle fails to detect oncoming/crossing vehicle and does not yield	L-2, L-3	Medium	High
H-4	Autonomous mode remains active during ODD violation (e.g. night, rain)	L-1, L-2, L-3	High	High
H-5	Vehicle brakes unnecessarily at high frequency due to false positives	L-2, L-3	Medium	Low

2.5 Control Structure Diagram

Draw the control structure for the CARLA system. Your diagram must include:
• All components listed in the system description (camera, perception models, distance oracle, planning module, actuators, human operator).
• Control actions (solid arrows): commands flowing from controllers to controlled processes (e.g., "brake command").

• Feedback (dashed arrows): information flowing back (e.g., “perception labels”, “distance estimates”, “vehicle speed”).

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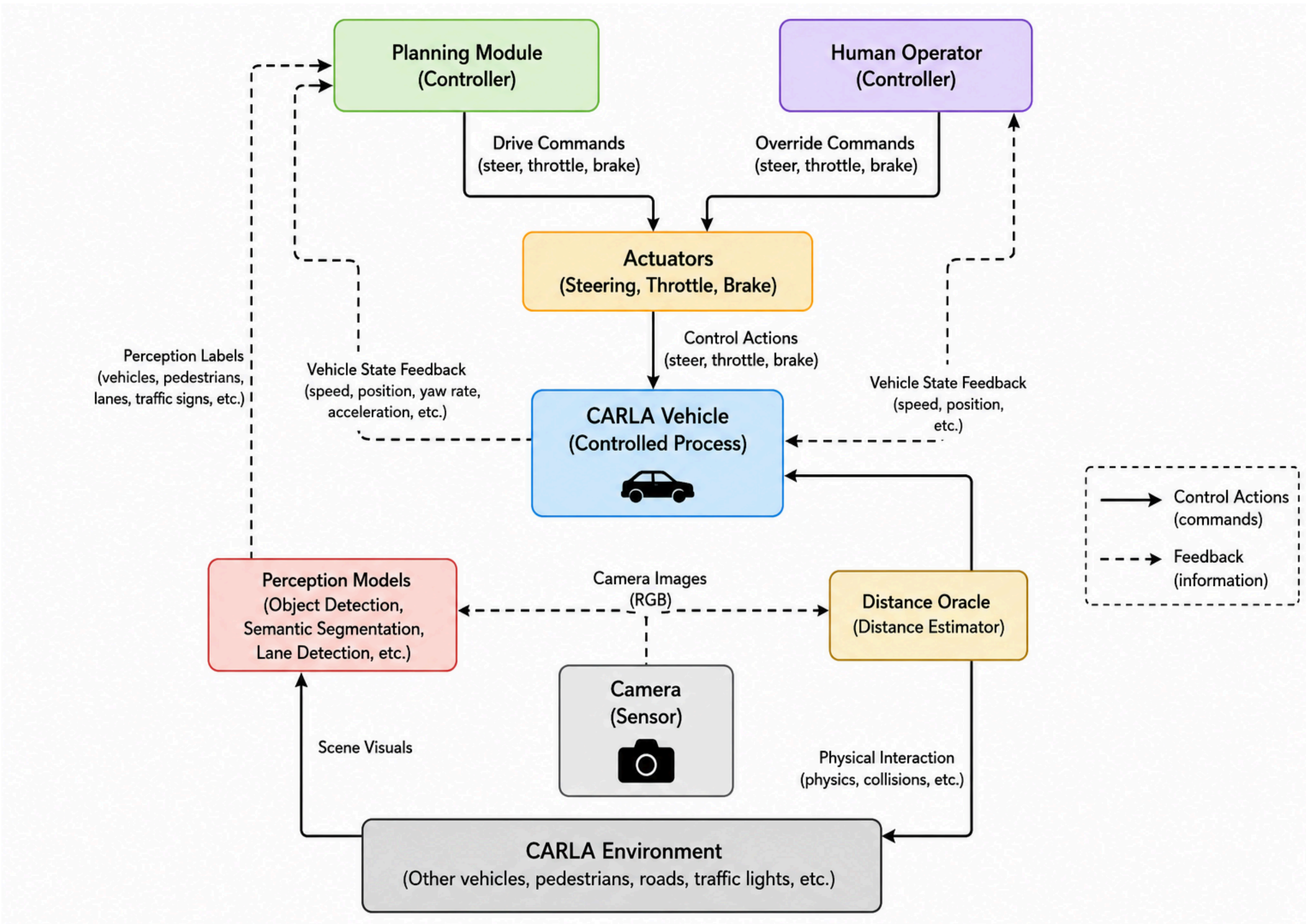

2.6 Unsafe Control Actions (UCAs)

Using your control structure, identify unsafe control actions for at least two different controllers. For each UCA, also indicate which hazard(s) from Exercise 2.4 it could cause or lead to (reference them by ID, e.g., “H-1”, “H-2”).

For each controller, consider all four UCA types:

1. Not provided - a required action is missing.
2. Provided unsafely - the action is issued when it should not be.
3. Wrong timing / order - too early, too late, or out of sequence.
4. Wrong duration - applied too long or stopped too early

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ID	Controller	Control Action	UCA Type	Hazard(s)	Unsafe Scenario
UCA-1	Planning Module	Brake command	Not Provided	H-1	The planning module does not issue a brake command when a pedestrian is detected in the vehicle's path, causing the vehicle to continue moving toward the pedestrian.
UCA-2	Planning Module	Stop command at traffic signal	Not Provided	H-2	The planning module fails to command the vehicle to stop at an intersection because the traffic light is not detected.
UCA-3	Planning Module	Yield / Brake command	Not Provided	H-3	The planning module does not issue a yield or brake command when another vehicle is crossing the intersection.
UCA-4	Planning Module	Disengage autonomous mode	Not Provided	H-4	The planning module does not deactivate autonomous mode when the vehicle enters rain or darkness outside the operational design domain (ODD).
UCA-5	Planning Module	Acceleration command	Provided Unsafely	H-1	The planning module commands acceleration while a pedestrian is directly in front of the vehicle.
UCA-6	Planning Module	Brake command	Provided Unsafely	H-5	The planning module repeatedly issues brake commands due to false positive detections, causing unnecessary frequent braking.

UCA-7	Planning Module	Brake command	Wrong Timing / Order	H-1, H-3	The brake command is issued too late after detecting a pedestrian or crossing vehicle, leaving insufficient stopping distance.
UCA-8	Planning Module	Brake command	Wrong Duration	H-1	Braking is released too early before the pedestrian has completely cleared the vehicle's path.
UCA-9	Human Operator	Manual override / Takeover	Not Provided	H-1, H-4	The human operator does not intervene when the autonomous system fails to brake or continues operating outside the ODD.
UCA-10	Human Operator	Override brake command	Provided Unsafely	H-1, H-3	The operator presses the accelerator or steers incorrectly instead of braking during a hazardous situation.
UCA-11	Human Operator	Manual override / Takeover	Wrong Timing / Order	H-1, H-2, H-3	The operator takes over too late to prevent a collision or intersection violation.
UCA-12	Human Operator	Return control to autonomous mode	Wrong Duration	H-1, H-4	The operator relinquishes control before the hazardous condition has been resolved.

2.7 Safety Constraints

For each UCA from Exercise 2.6, derive at least one safety constraint - a requirement that, if satisfied, prevents the UCA from occurring or limits its consequences.
Classify each constraint as either:
• Model-level - can be verified by testing or evaluating the corresponding perception model (this is what you will do in Sheets 4–9).
• System-level - requires a design measure beyond the model (e.g., redundancy, operator protocol, fallback behavior).

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UCA	Safety Constraint	Level	Verification
UCA-1: Planning module does not issue a brake command when a pedestrian is detected (H-1)	The planning module shall issue a brake command whenever the perception system detects a pedestrian within the stopping distance in the vehicle's path.	Model-level	Evaluate pedestrian detection recall and distance estimation accuracy on test scenarios with pedestrians directly ahead.
UCA-2: Planning module fails to stop at an undetected traffic signal (H-2)	The system shall detect and correctly classify traffic lights and command a stop when the signal is red.	Model-level	Test traffic light detection and classification accuracy under varied intersection scenarios.
UCA-3: Planning module fails to yield to crossing vehicles (H-3)	The system shall detect crossing vehicles and issue braking or yielding commands when a collision risk exists.	Model-level	Measure vehicle detection recall and time-to-collision estimation performance.
UCA-4: Autonomous mode remains active outside the ODD (H-4)	The system shall monitor environmental conditions (e.g., rain, darkness) and automatically disengage autonomous mode or request operator takeover when ODD limits are exceeded.	System-level	Validate fallback behavior in simulation by introducing rain/night conditions and checking takeover requests.
UCA-5: Planning module accelerates toward a pedestrian (H-1)	The planning module shall inhibit acceleration commands whenever an obstacle is detected within the minimum safe stopping distance.	System-level	Perform integration tests to confirm acceleration is blocked under close-obstacle conditions.
UCA-6: Repeated false-positive braking (H-5)	The system shall require confidence thresholds and temporal consistency before issuing emergency braking commands.	Model-level	Evaluate false positive rate and confidence calibration of the perception model.

UCA-7: Brake command issued too late (H-1, H-3)	The system shall issue braking commands within a maximum allowable reaction time after hazard detection.	System-level	Measure end-to-end latency from perception output to actuator command in simulation.
UCA-8: Braking released too early (H-1)	The system shall maintain braking until the obstacle is no longer within the collision zone.	System-level	Simulate pedestrian crossing scenarios and verify braking duration.
UCA-9: Human operator does not take over when required (H-1, H-4)	The system shall provide timely and escalating takeover alerts when autonomous control becomes unsafe.	System-level	Conduct human-in-the-loop simulations and measure operator response times.
UCA-10: Operator overrides unsafely (H-1, H-3)	The system shall validate manual override commands and suppress commands that would immediately increase collision risk.	System-level	Test override logic by injecting unsafe operator inputs in simulation.
UCA-11: Operator takes over too late (H-1, H-2, H-3)	The system shall issue takeover requests with sufficient lead time to allow safe intervention.	System-level	Measure warning lead time and compare with human reaction benchmarks.
UCA-12: Operator returns control too early (H-1, H-4)	Autonomous mode shall not be re-engaged until hazards are cleared and ODD conditions are satisfied.	System-level	Verify re-engagement logic in scenarios involving temporary hazards and ODD violations.

2.8 Causal Loss Scenarios

For each UCA from Exercise 2.6, identify at least one causal scenario - a specific mechanism or sequence of events that could cause the UCA to occur in practice.

For each scenario, describe:

1. The causal factor: What component failure, misunderstanding, or adverse condition triggers the UCA?
2. Why it occurs: What is the root cause or underlying reason?
3. Connection to constraint: Which safety constraint from Exercise 2.7 is designed to prevent or mitigate this scenario?

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UCA	Causal Factor	Why It Occurs (Root Cause)	Related Safety Constraint
UCA-1: Planning module does not issue a brake command when a pedestrian is detected	Pedestrian is missed by the perception model	Insufficient training data, occlusion, poor lighting, or low model recall causes the pedestrian not to be detected	SC-1: The planning module shall issue a brake command whenever a pedestrian is detected within stopping distance
UCA-2: Planning module fails to stop at a red traffic signal	Traffic light is not detected or misclassified	Small object size, unusual angle, weather effects, or classification errors	SC-2: The system shall detect and correctly classify traffic lights and command a stop when the signal is red
UCA-3: Planning module fails to yield to a crossing vehicle	Crossing vehicle is not detected or its trajectory is mispredicted	Occlusion, sensor limitations, or inaccurate time-to-collision estimation	SC-3: The system shall detect crossing vehicles and issue braking or yielding commands when collision risk exists
UCA-4: Autonomous mode remains active outside the ODD	ODD monitoring fails to recognize rain or darkness	Missing environmental sensors, incorrect thresholds, or software logic errors	SC-4: The system shall automatically disengage autonomous mode or request takeover when ODD limits are exceeded
UCA-5: Planning module accelerates toward a pedestrian	Incorrect perception output labels the path as clear	False negative detection or stale sensor data causes the planner to believe acceleration is safe	SC-5: The planning module shall inhibit acceleration when an obstacle is within safe stopping distance

UCA-6: Repeated false-positive braking	Perception model incorrectly detects obstacles	Poor calibration, noisy images, or low confidence thresholds generate false positives	SC-6: The system shall require confidence thresholds and temporal consistency before emergency braking
UCA-7: Brake command is issued too late	Processing or communication latency delays control	High computational load, slow model inference, or delayed actuator response	SC-7: The system shall issue braking commands within a maximum allowable reaction time
UCA-8: Braking is released too early	Distance estimate incorrectly indicates the obstacle is gone	Noisy distance measurements or premature state transitions in the planner	SC-8: The system shall maintain braking until the obstacle is outside the collision zone
UCA-9: Human operator does not take over when required	Operator is distracted or unaware of takeover request	Poor alert design, fatigue, or insufficient training	SC-9: The system shall provide timely and escalating takeover alerts
UCA-10: Operator overrides unsafely	Operator misjudges the situation and presses the accelerator	Stress, misunderstanding of vehicle state, or poor situational awareness	SC-10: The system shall suppress override commands that would immediately increase collision risk
UCA-11: Operator takes over too late	Takeover request is issued with insufficient warning	Late hazard detection or inadequate alert timing	SC-11: The system shall issue takeover requests with sufficient lead time
UCA-12: Operator returns control to autonomous mode too early	Hazard persists or ODD conditions are still violated	Operator assumes the danger has passed or system provides unclear status information	SC-12: Autonomous mode shall not be re-engaged until hazards are cleared and ODD conditions are satisfied